

TAHA KOULAL

SOFTWARE ENGINEER | BACKEND, AI & PRODUCT ENGINEERING

tkoulal.21@gmail.com | +212 646 564 974 | github.com/tahariginal | Benguerir, Morocco

PROFILE

Product-minded software engineer with a systems foundation and hands-on experience building backend services, real-time applications, AI-agent workflows, and web and mobile products from zero.

Strong at translating ambiguous business needs into secure architecture, practical workflows, and maintainable delivery systems.

TECHNICAL STACK

Backend & Systems
C, C++, Unix/Linux, TCP/IP, REST APIs, WebSockets, PostgreSQL, Supabase

Product Engineering
TypeScript, JavaScript, React, React Native, Next.js, real-time systems

AI & Automation
LLM agents, Claude API, MCP, prompt engineering, n8n, tool integration

ENGINEERING STRENGTHS

- 0-to-1 product development
- Architecture and technical scoping
- RBAC, security and data boundaries
- Workflow and systems design
- Cross-functional collaboration
- AI-assisted delivery with review gates

EDUCATION

42 NETWORK (1337)
Software Engineering
Common Core in Progress

Validated ft_irc, Minishell, Philosophers, ft_printf, Libft, and 10+ projects.

UM6P - IDC Morocco
VR / Interactive Technologies

LANGUAGES

Arabic - Native
English - Professional
French - Intermediate

SELECTED ENGINEERING WORK

Factory | Multi-Agent AI Project Factory

Claude Code • MCP • Agent orchestration

- Architected an agent-based delivery system where a CEO agent evaluates ideas, delegates execution to specialised CTO modes, and coordinates focused build agents.
- Designed permission gates, milestone reviews, persistent state, audit logs, sandboxed execution, and SLA-based blocker escalation.
- Modelled the system around small, specialised agent teams to improve reliability, accountability, and deterministic project execution.

DPRO | Business Opportunity Platform

PHOVA Technology • Next.js • Supabase

- Contributed to a production platform managing opportunities across sales, technical evaluation, costing, sourcing, and delivery.
- Built and improved RBAC, notifications, workflow rules, opportunity management, and complex technical interfaces with business stakeholders.
- Simplified multi-step technical screens by removing duplicated information and aligning the interface with real operational responsibilities.

Vamo | Football Identity Mobile Product

React Native • Supabase • Product engineering

- Designed and built a mobile product for persistent player identities, matches, teams, tournaments, discovery, history, and a Football Passport.
- Implemented privacy-aware Supabase RLS, including teammate-only phone visibility and secure player-data access.
- Defined the product structure and core flows from zero, including discovery, match and tournament creation, attendance, and profile history.

ft_transcendence | Real-Time Multiplayer Platform

React • TypeScript • WebSockets

- Built authentication, live messaging, synchronised game state, and event-driven communication for a multiplayer Pong platform.
- Worked on real-time frontend state and game-session behaviour supporting more than 20 concurrent games.
- Applied event-driven communication patterns to keep multiplayer state, chat, and user interactions synchronised.

FIT | AI Wardrobe and Outfit Mobile App

Product concept • React Native • Recommendation system

- Designed a product concept that converts a user's existing wardrobe into daily outfit recommendations rather than unnecessary purchases.
- Structured recommendations around wardrobe items, weather, skin tone, body shape, colour compatibility, and personal preferences.
- Positioned the experience around wardrobe reuse: users catalogue what they own once, then receive context-aware complete looks each day.

EXPERIENCE

PHOVA Technology | Software Engineering Intern

Delivered backend and product features inside an existing codebase and translated operational requirements across sales, engineering, sourcing, and management into usable workflows.

Worked across backend behaviour, access control, notifications, workflows, and product UX while coordinating feedback with non-technical stakeholders.

UM6P - IDC Morocco | VR Development Intern

Developed an immersive virtual-visit experience from prototype to functional delivery using Unity and C#, including interactive 3D environments and navigation.

ENGINEERING APPROACH

Start with the user and business workflow, translate ambiguity into explicit priorities, then build secure foundations with clear permissions, data boundaries, auditability, and review gates. Use AI to accelerate delivery while keeping engineering judgment and product quality central.

Available for full-time and early-stage product opportunities